Oral Exam Guidance

Quantitative Reasoning II · Final Project

The oral exam is a 15-minute, one-on-one meeting about your project, scheduled individually during finals week. This document explains the format and focus of the meeting.

1 Resources

You may not use resources during the exam. I'll bring a printed copy of your report and your notebook for us both to reference. You don't have to have exact quantities memorized if you know where to find them; "let's look at Table 2, column 3" is a fine way to begin an answer.

2 Format

The exam has three parts.

Opening: everyone will have the same two opening questions:

- In a sentence or two: why did you pick this question?
- What does one row of your data represent?

Middle: questions written based on your submission, drawing from the topics listed below. Each topic starts with a base question and may escalate in difficulty depending on your answers. How many topics we reach depends on the clock and on how far each one goes, so be ready for any topic that applies to your project.

Closing: everyone will have the same closing prompt:

- briefly, what should someone (who hasn't necessarily taken a statistics class) take away from your project?

3 Ground rules, what to expect

The oral exam is like an interview. This will probably feel a little uncomfortable! We only have fifteen minutes, and the goal is to assess your understanding and communication of the statistical concepts in your project so I won't speak much. To be fair to all students, I won't be able to give feedback on your performance live. I'll mark my exam sheet as we go, regardless of your performance (don't take my notes as an indicator that you got something wrong). Grades will be released after all exams finish. Some notes:

- I won't deliver feedback live: I won't affirm correct answers or correct mistakes.
- Follow-up questions are not a signal of performance: correct and incorrect answers may both receive follow-ups.
- If you don't know, say you don't know and we will move on.
- I'll run a timer where I can see it and pivot to the closing prompt with about one minute left in the allotted time.

4 Topics

Which topics you receive depends on the tools you use in your analysis. Everything in your report is fair game: don't include a table or figure if you can't speak to it! Below are the topics you should be ready to discuss, with a non-comprehensive list of the concepts in each topic.

Topic	Example concepts
Causal inference	association versus cause, confounders, what you held constant, what you're not able to control for
Regression basics and interpretation	coefficients, R^2, why an estimate moved when controls went in
Uncertainty	p-values, confidence intervals, how much your estimate might move
Distributions and measurement	what your variables count, how well your measures capture the thing you're interested in, the shape and spread of your data
Selection	how your data was collected, what your sample represents, what your sample doesn't include
Limitations and strength of evidence	your biggest limitations, strengths of your analysis

5 Preparation

The questions are related to your work, so preparation is more about understanding and communicating your work rather than memorizing answers to questions. Some things you may do as you finish your project and prepare for your exam:

- Review the midterm study guide and course slides.
- Reread your report (including tables and figures) until you can navigate them easily.
- Guide a friend or roommate through your report and answer their questions.
- Check that the numbers in your report match your figures and tables.
- Check that your notebook runs top to bottom and computes your tables and figures *from your data*, not from numbers typed in by hand.
- Articulate your reasons for all of the choices you made in your analysis (filters, variables, model type, etc.).
- Ask a model to interview you about your project — give it your report, notebook, and our course materials as context. Ask it to give you follow-up questions and to push back if your answer is vague or incorrect.

6 Grading

The bulk of the grade comes from the middle section of the exam (your project-specific questions). The grade will consider both the depth and breadth of your understanding and the clarity of your answers. Be confident in your abilities, but don't be afraid to say you don't know when we reach your limit.